Johns Hopkins University Data Management Services: Reviewing Our First Year

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Our service development: a social science analogy

Developing new services for data management support required understanding social factors, not just technical.

- Services emerging from data curation
  "social movement"
- Investigating
  - **University culture** for outreach
  - **Data practices** for consulting & curation support
  - **Trends** in public funding, data sharing & curation

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Data curation "social movement"

Emergence — Recognizing the need for curation & access

- eScience: digital data production outpaces sharing & preservation

Coalescence — Policy & requirements

- Jan 2011: NSF Data Management Plan requirement

Institutionalization — Building Services & Infrastructure

- 2009: NSF DataNet funds Data Conservancy at JHU

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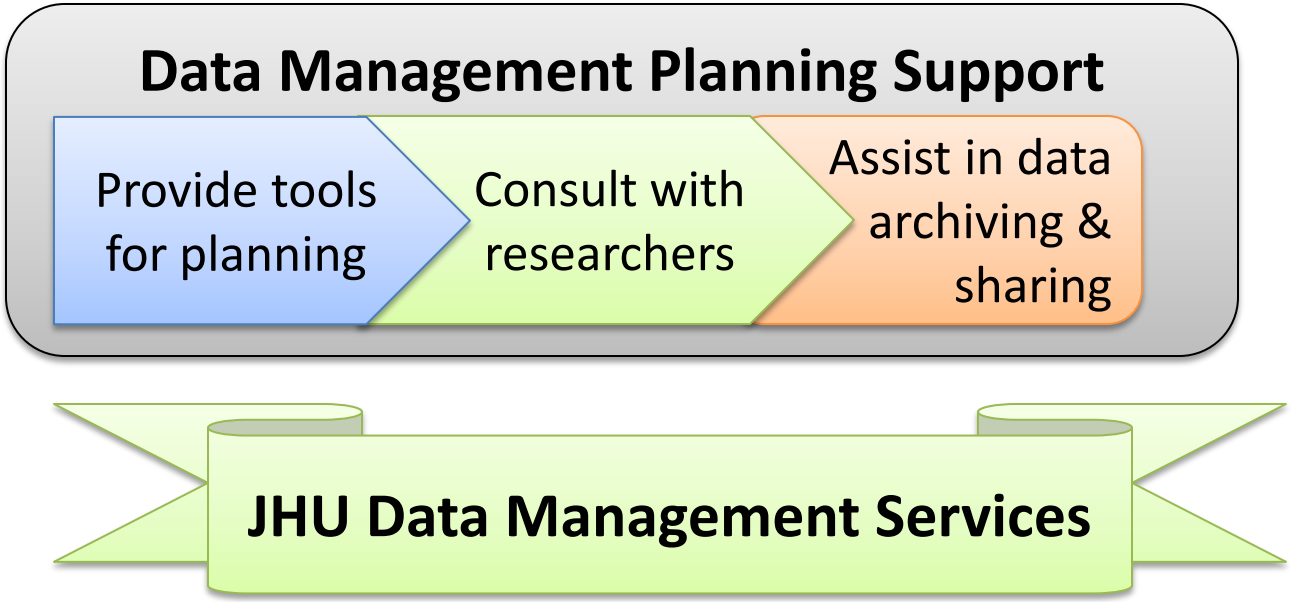
Data Management Services at Academic Libraries

- Academic libraries in early stages of expanding services to support data curation & management



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JHU Data Management Services

	Preparing Data Management Plans	Archiving Research Data
Services	Assistance with data management plans	Assist in implementing plan, preserve & share research through JHU Data Archive
Cost	Funded by JHU schools using our service	2% of grant's total direct cost

- JHU Data Archive features (in development):

Data Conservancy architecture

Accepts data from all disciplines

Feature-level discovery

Interoperable data access

Persistent identifiers

Supports long-term preservation

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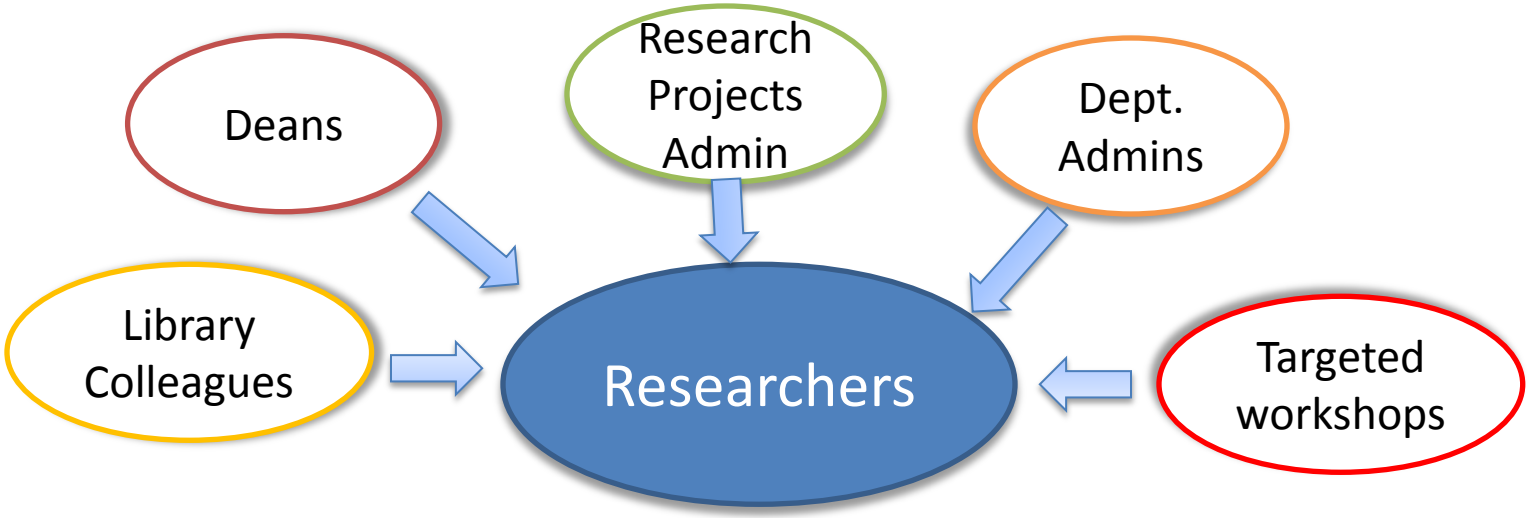
Investigating JHU Cultures to introduce Data Management Services

Outreach Goals	Approach
Building awareness of services	Finding appropriate communication channels



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Investigating JHU Cultures to introduce Data Management Services

Outreach Goals	Approach
Building awareness of services	Finding appropriate communication channels
Demonstrating Relevance of services	Consulting and customized support

Do DMPs matter to NSF?

Can't I copy a template?

Why archive my data?

Investigating data practices

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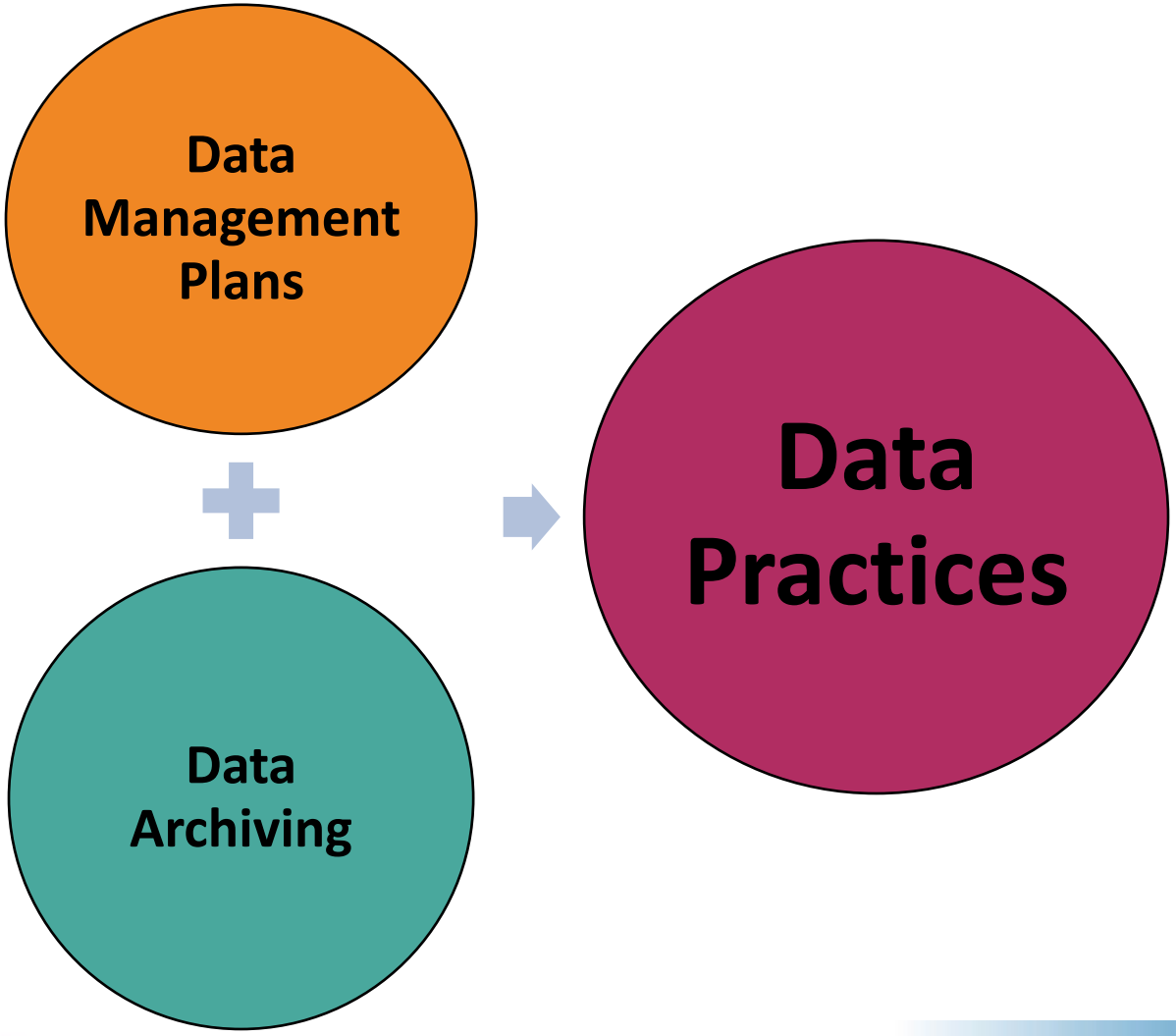
Understanding Data Practices



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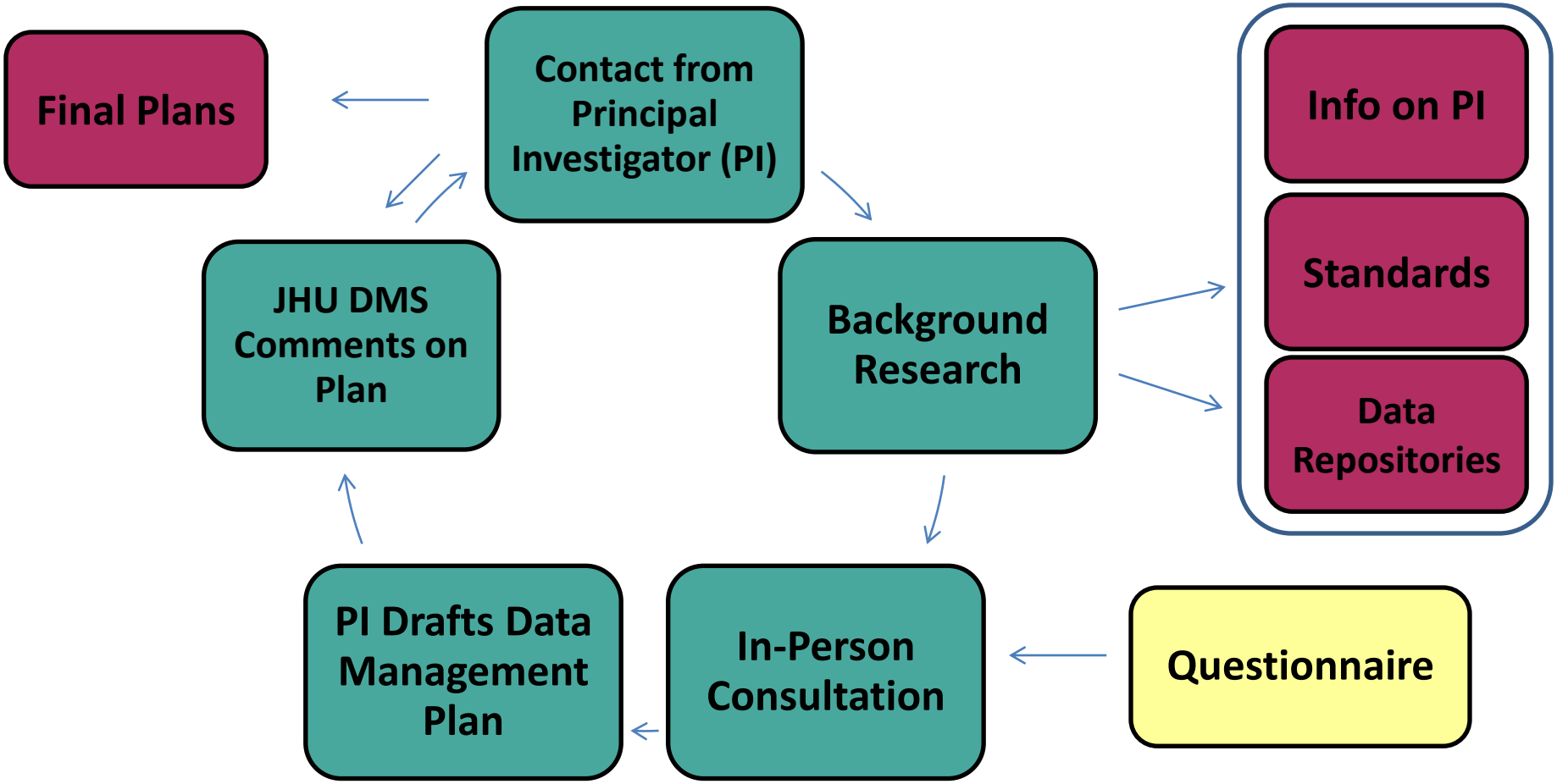
Consulting on data management plans



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In-Person Consultation Tool:

Questionnaire (sample questions)

Section I: Data Products and Standards
What naming conventions/schema will be used for your data, if any?
Section II: Data Storing and Long-Term Preservation
During the project, how (i.e. media) and where (i.e. location(s)) will data be stored and who is responsible for it?
Section III: Data Sharing
Are there any data with privacy concerns to sharing (e.g., human subjects)? If so, what policies need to be adhered to and how will policies be enforced?

Social vs. Physical Science:

Observations from data plans

Data Type/ Qty	• Mixture of digital and physical material:
	  • digital (documents, audio recordings, photos, database)
	  • non digital (newspaper, interview notes, artifacts)
	• Datasets on the scale of MBs and GBs, not TBs

Personal Identifiers	• Use of human subjects does not exempt one from sharing data ○ Graduate student received grant but was asked to modify data management plan and find something to share
Metadata	• More familiar with term “codebook” than “metadata” • Metadata is more often created manually by the PI

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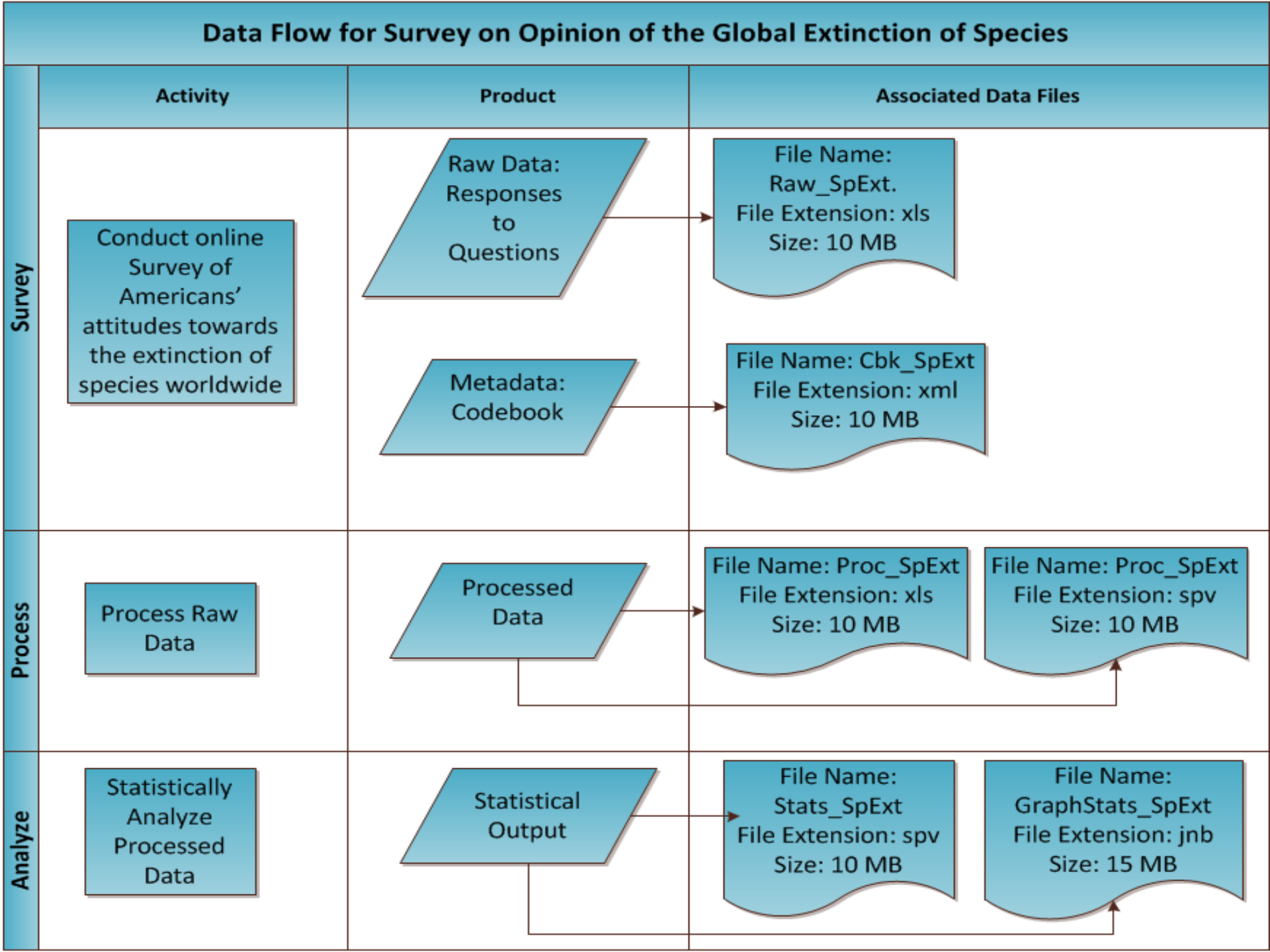
Data Flow for Survey on Opinion of the Global Extinction of Species
Activity
Product
Associated Data Files
Survey
Conduct online Survey of Americans' attitudes towards the extinction of species worldwide
Raw Data: Responses to Questions
File Name: Raw_SpExt. File Extension: xls Size: 10 MB
Metadata: Codebook
File Name: Cbk_SpExt File Extension: xml Size: 10 MB
Process
Process Raw Data
Processed Data
File Name: Proc_SpExt File Extension: xls Size: 10 MB
File Name: Proc_SpExt File Extension: spv Size: 10 MB
Analyze
Statistically Analyze Processed Data
Statistical Output
File Name: Stats_SpExt File Extension: spv Size: 10 MB
File Name: GraphStats_SpExt File Extension: jnb Size: 15 MB

Investigating trends in data curation

Funders new requirements & policy

- Other funders adopting data management plan requirements

Data curation archiving, repositories, standards

- Expanding our expertise and knowledgebase

Data sharing Collaboration, data-intensive science

- E.g., academic credit for sharing data collections?

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Thank you

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